SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA0706 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across

networks. (BL2, BL3)
Assessment Tool Weightage: 30%

Annexure A
CONCEPT MAPPING

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Code of Conduct:

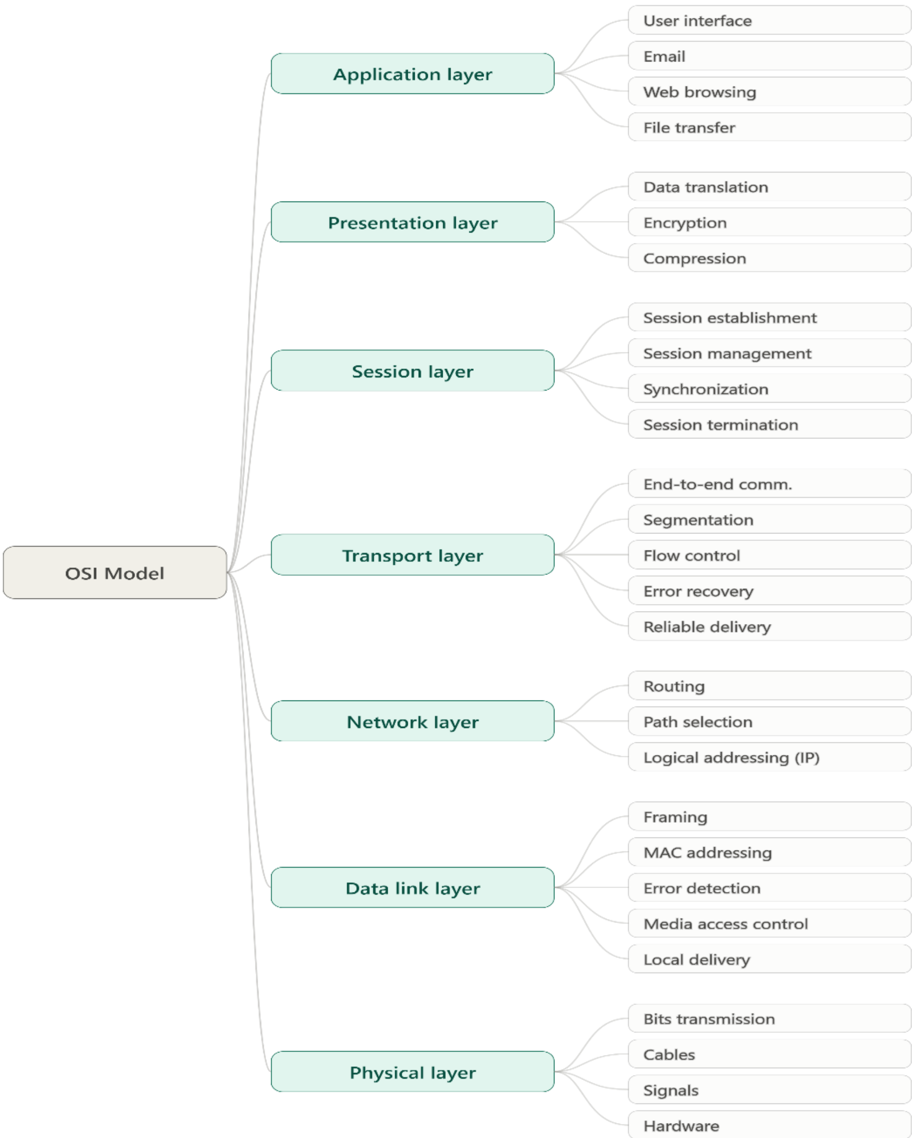
I D.Uday Kiran (192425161)**(Name / Reg No) certify that this submission is my original work**

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:D.Uday Kiran

Question-1
OSI Layer
Concept Map



Missing Layers

- **Session Layer**
- **Data Link Layer**

Explanation-Session Layer

- Establishes, manages, and terminates communication sessions.
- Maintains synchronization between sender and receiver.
- Supports session recovery after interruptions.

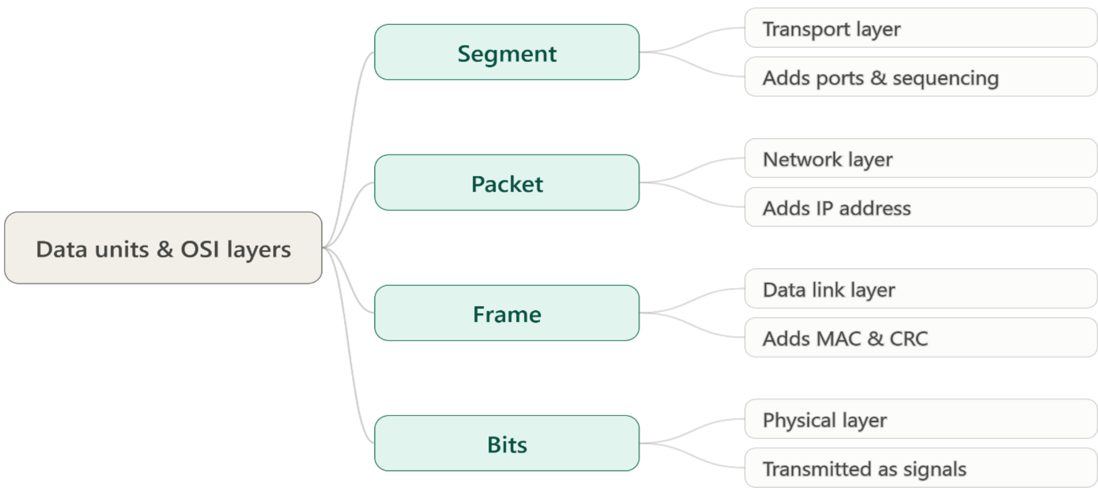
Data Link Layer

- Provides node-to-node communication.
- Detects and corrects transmission errors.
- Uses MAC addresses for local delivery.
- Controls data flow and framing.

Analysis

Reliable communication depends on every OSI layer. The Session layer maintains active communication, while the Data Link layer ensures error-free local transmission before data reaches the Physical layer.

Question 2 Data Units and Corresponding OSI Layers Concept Map



Explanation

Data Unit	OSI Layer	Function
Segment	Transport	End-to-end communication
Packet	Network	Routing using IP address
Frame	Data Link	Local delivery using MAC address
Bits	Physical	Electrical/Optical signal transmission

Analysis

During encapsulation, application data becomes a Segment, then Packet, then Frame, and finally Bits. At the receiver, decapsulation converts Bits back into the original message.

How Transformation Supports Transmission

- Transport Layer divides data into segments.
- Network Layer adds logical addressing and creates packets.
- Data Link Layer creates frames and adds error-checking information.

• Physical Layer converts frames into binary bits for transmission.

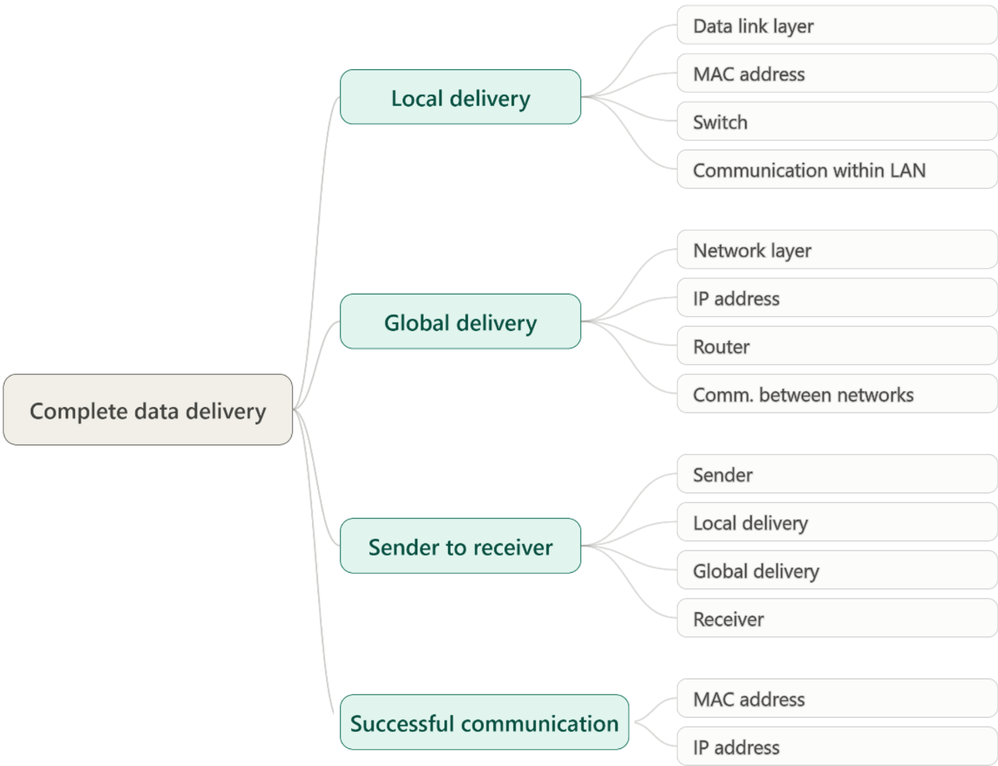
This process is known as **Encapsulation**.

Conclusion

Transformation of data units allows data to move efficiently and reliably through the network

Question 3

MAC Address and IP Address Relationship Concept Map



Analysis

MAC Address

• Physical address of a device.

• Used within the local network.

• Ensures delivery between devices on the same LAN.

IP Address

• Logical address assigned to devices.

- Used for communication across multiple networks.
- Enables routing over the Internet.

Importance

Both addresses work together:

- IP address identifies the destination network.
- MAC address identifies the destination device.

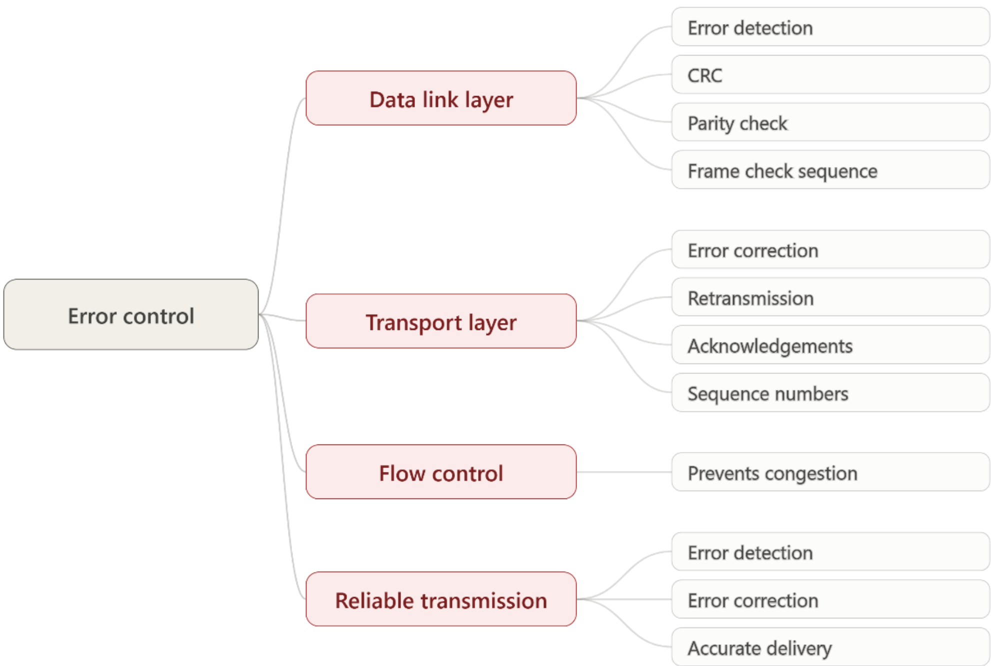
Conclusion

The combination of MAC and IP addressing ensures successful data delivery from source to destination across local and global networks

Question 4

Error Control

Mechanisms Concept Map



Error Correction

Analysis

Data Link Layer

- Detects transmission errors.

- Uses CRC and checksum methods.
- Identifies corrupted frames.

Transport Layer

- Performs end-to-end reliability.
- Uses acknowledgments and retransmissions.
- Corrects errors by resending lost data.

Reliability Improvement

Error detection identifies problems quickly, while error correction ensures lost or damaged data is recovered.

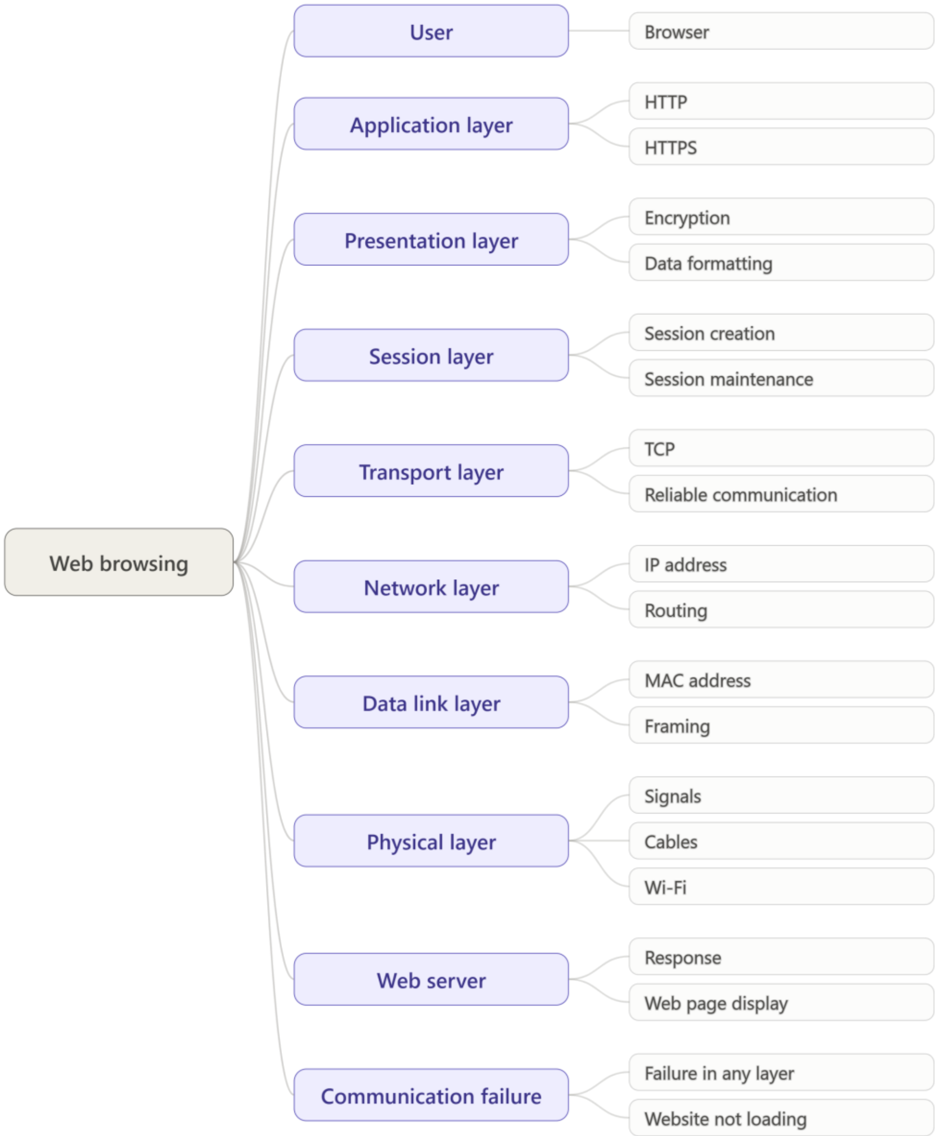
Conclusion

The combined operation of these layers provides highly reliable communication.

Question 5

Web Browsing

Concept Map



Analysis

When a user opens a website:

1. Application Layer sends an HTTP request.
2. Presentation Layer encrypts data using SSL/TLS.
3. Session Layer maintains the connection.
4. Transport Layer ensures reliable delivery using TCP.
5. Network Layer routes packets.
6. Data Link Layer creates frames.
7. Physical Layer transmits bits through cables or wireless signals.

Failed Layer Effect

Application Website cannot load

Presentation Data cannot be encrypted/decrypted Session Connection interruptions

Transport Packet loss and retransmissions Network Routing failure

Data Link Frame transmission errors

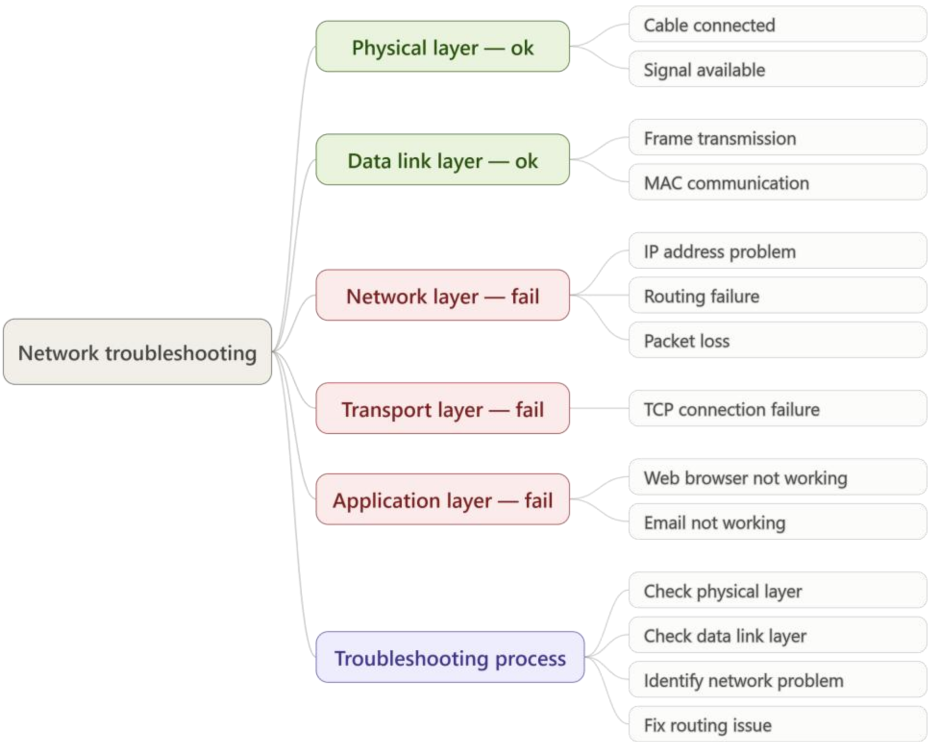
Physical No network connectivity

Conclusion

All OSI layers depend on one another; failure of one layer affects the entire communication process.

Question 6

Network Troubleshooting Using Concept Mapping



Analysis

The concept map indicates:

- Physical Layer is functioning correctly.
- Data Link Layer is functioning correctly.
- Network Layer has failed.

Because the Network Layer is not working:

- IP routing cannot occur.
- Packets cannot reach the destination.
- Transport Layer cannot establish communication.
- Applications cannot exchange data.

Possible Causes

- Incorrect IP address
- Routing table errors
- Router failure
- Network configuration issues

How Concept Mapping Helps

Concept maps show dependencies between layers. By identifying the first failed layer, network administrators can quickly locate and resolve problems.

Conclusion

Since the Network Layer fails, all upper layers also fail. Concept mapping simplifies troubleshooting and reduces diagnostic time.

Overall Conclusion

The OSI model provides a structured framework for network communication. Concept mapping helps visualize interactions among layers, addresses, data units, error-control mechanisms, and troubleshooting procedures. Understanding these relationships improves network design, maintenance, reliability, and performance.